speeds, momentarily produce tiny fireballs so hot and dense that they smush together all the particles and momentarily fill the collider with a quark-gluon plasma state. By watching the debris “freeze” out into ordinary hadrons, scientists can study the properties of this exotic matter as well as the way the laws of physics act under those extreme conditions.

If seeing the CMB gives us a glimpse of the Big Bang, high-energy particle colliders are giving us a taste of the primordial soup.[fn16]

BIG BANG NUCLEOSYNTHESIS

After the quark-gluon plasma phase, the universe started to cool down enough for some familiar particles to form. At around a tenth of a millisecond, the first protons and neutrons formed, followed shortly by electrons, laying down the building blocks of ordinary matter. Somewhere around the two-minute mark, the universe cooled to a comfortable billion degrees Celsius, hotter than the center of the Sun but cool enough to allow the strong force to bring those shiny new protons and neutrons together. They formed the first bonded atomic nucleus: a form of hydrogen called deuterium (one proton bonded to one neutron; technically a single proton can be considered a nucleus too, since it’s the center of a hydrogen atom). Soon, nuclei were forming left and right. Some fraction of the protons and neutrons started joining together to make helium nuclei, tritium, and a smattering of lithium and beryllium. This process, called Big Bang Nucleosynthesis, went on for about half an hour, until the universe cooled down and expanded enough that the particles were able to get away from each other instead of fusing together.

One of the great validations of the Big Bang theory is the fact that we find a close match between our observations in the cosmos and the calculated abundance of elements we expect from the Big Bang based on our estimates of the temperature and density of that primordial fireball. The agreement isn’t perfect—there’s some lingering confusion around the lithium abundance that may or may not be telling us about some extra weirdness going on in the early universe—but with hydrogen, deuterium, and helium, measuring how much we actually see out there and comparing it to what we calculate *should* happen if you shove the entire cosmos into a nuclear furnace results in some absolutely beautiful concordance.